

Students will describe and perform reflections across given lines of reflection.

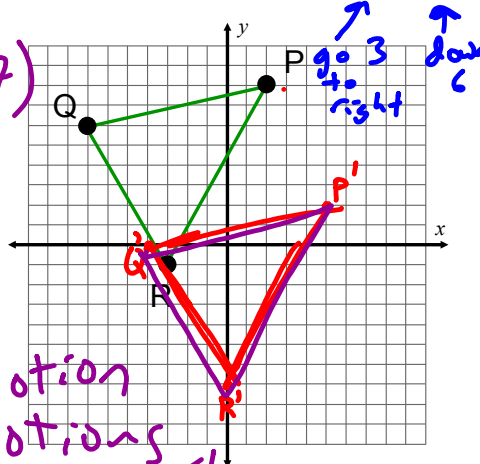
Do Now

Triangle PQR has coordinates $P(2, 8)$, $Q(-7, 6)$ and $R(-3, -1)$.

a. On the set of axes below, graph, label, and state the coordinates of $\Delta P'Q'R'$, the image of ΔPQR under the translation $(x, y) \rightarrow (x + 3, y - 6)$.

$Q'(-4, 0)$
 $P'(5, 2)$

$R'(0, -7)$



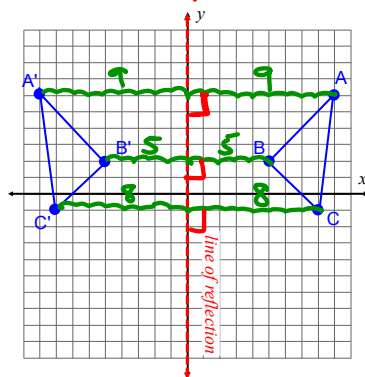
b. Explain why ΔPQR is congruent to $\Delta P'Q'R'$.

A translation is a rigid motion and rigid motions preserve side lengths and angle measures.

Reflections

A reflection is a rigid motion that sends each point in a figure to a point of equal distance on the opposite side of a given line, called the line of reflection, or across a point, called the point of reflection.

1) Fill in the coordinates for each of the following reflections, then make a rule.



$A(9, 6) \rightarrow A'(-9, 6)$

$B(5, 2) \rightarrow B'(-5, 2)$

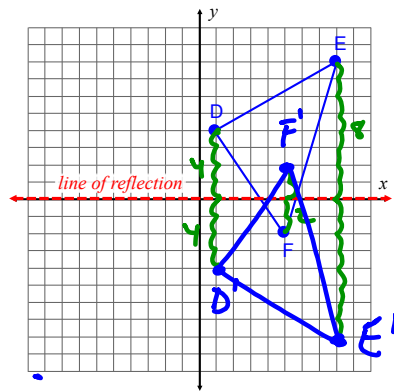
$C(8, -1) \rightarrow C'(-8, -1)$

Coordinate rule for y-axis

$(x, y) \rightarrow (-x, y)$

"Switch the sign of x, keep y the same"

2) Fill in the coordinates for the pre-image shown below. Reflect each point across the line of reflection, and fill in the coordinates of the image.



Pre-image Image
 $D(\underline{1}, \underline{4}) \rightarrow D'(\underline{1}, \underline{-4})$

$E(\underline{8}, \underline{8}) \rightarrow E'(\underline{8}, \underline{-8})$

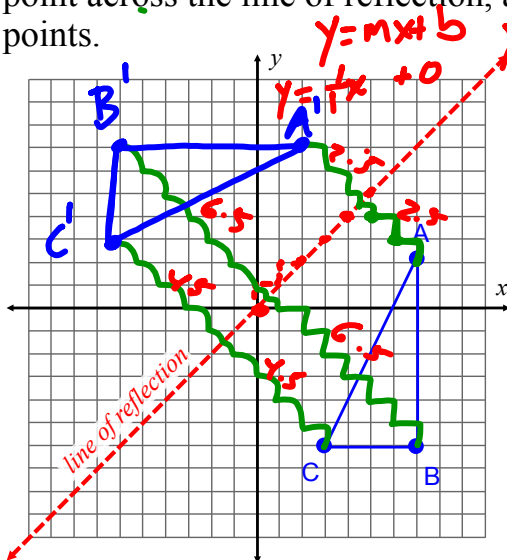
$F(\underline{5}, \underline{-2}) \rightarrow F'(\underline{5}, \underline{2})$

Coordinate rule for $r_{x\text{-axis}}$

$(x, y) \rightarrow (\underline{x}, \underline{-y})$

"Keep x the same, switch sign of y."

3) Fill in the coordinates for the pre-image shown below. Reflect each point across the line of reflection, and fill in the coordinates of the points.



Pre-image Image
 $A(\underline{7}, \underline{2}) \rightarrow A'(\underline{2}, \underline{7})$

$B(\underline{7}, \underline{-6}) \rightarrow B'(\underline{-6}, \underline{7})$

$C(\underline{3}, \underline{-6}) \rightarrow C'(\underline{-6}, \underline{3})$

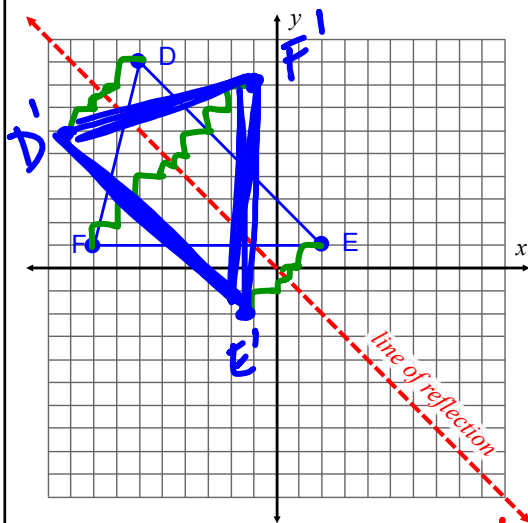
Coordinate rule for $r_{y=x}$

$(x, y) \rightarrow (\underline{y}, \underline{x})$

Switch x and y

Note: The line of reflection will always be the perpendicular bisector to the invisible line connecting two corresponding points.

4) Fill in the coordinates for the pre-image shown below. Reflect each point across the line of reflection, and fill in the coordinates of the



Pre-image

Image

$$D(\underline{-6}, \underline{9}) \longrightarrow D'(\underline{-9}, \underline{6})$$

$$E(\underline{2}, \underline{1}) \longrightarrow E'(\underline{-1}, \underline{-2})$$

$$F(\underline{-8}, \underline{1}) \longrightarrow F'(\underline{-1}, \underline{8})$$

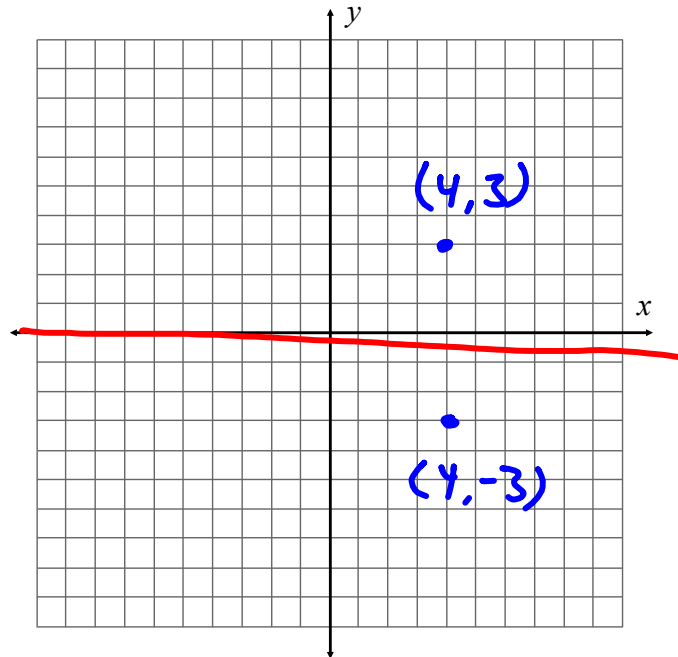
Coordinate rule for $r_{y=-x}$

$$(x, y) \longrightarrow (\underline{-y}, \underline{-x})$$

"switch positions of x and y and flip signs"

5) Complete the missing information in each reflection below. Feel free to use the coordinate grid shown below.

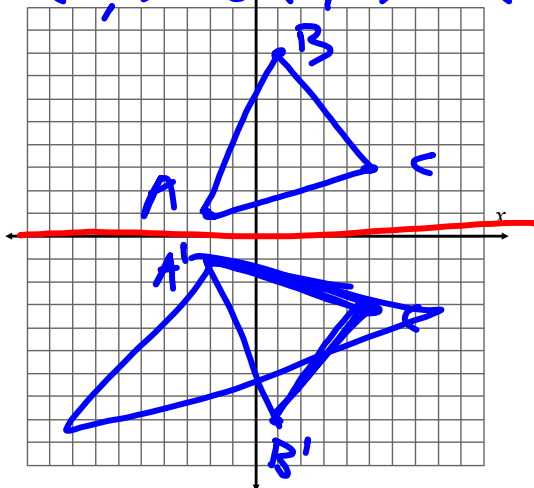
$r_{x\text{-axis}}(x, y) = (x, -y)$	$r_{x\text{-axis}}(4, 3) = (4, -3)$
$r_{x\text{-axis}}(-5, -2) = (-5, 2)$	$r_{x\text{-axis}}(7, -1) = (7, 1)$
$r_{y\text{-axis}}(x, y) = (-x, y)$	$r_{y\text{-axis}}(4, 3) = (-4, 3)$
$r_{y\text{-axis}}(-5, -2) = (5, -2)$	$r_{y\text{-axis}}(7, -1) = (-7, -1)$
$r_{y=x}(x, y) = (y, x)$	$r_{y=x}(4, 3) = (3, 4)$
$r_{y=x}(-5, -2) = (-2, -5)$	$r_{y=x}(7, -1) = (-1, 7)$
$r_{y=-x}(x, y) = (-y, -x)$	$r_{y=-x}(4, 3) = (-3, -4)$
$r_{y=-x}(-5, -2) = (2, 5)$	$r_{y=-x}(7, -1) = (1, -7)$



Practice

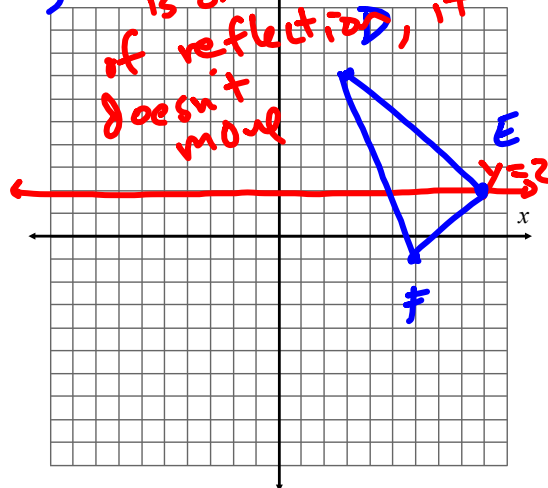
1) Triangle ABC has vertices $A(1,8)$, $B(5,3)$, and $C(-2,1)$. Find the coordinates of the vertices of $A'B'C'$ after the reflection $r_{x\text{-axis}}$.

$A'(1,-8)$ $B'(5,-3)$ $C'(-2,-1)$

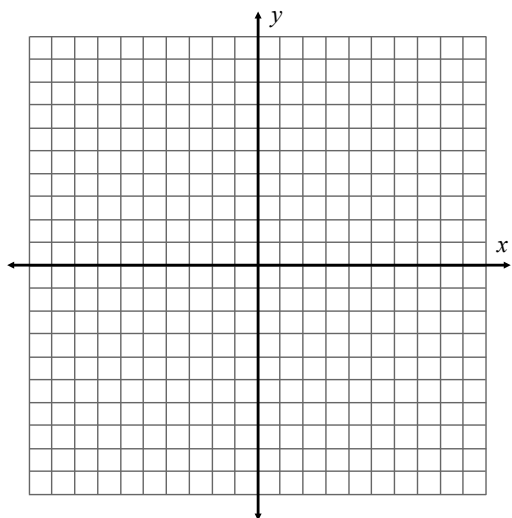


2) Triangle DEF has vertices $D(3,7)$, $E(9,2)$, and $F(6,-1)$. Find the coordinates of the vertices of $D'E'F'$ after the line $y=2$.

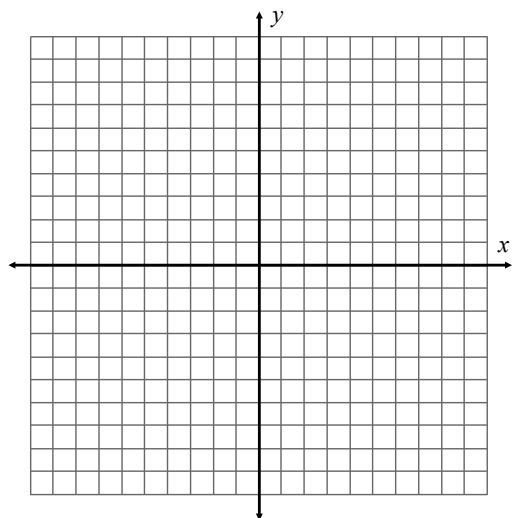
* If the point is on the line, it doesn't move



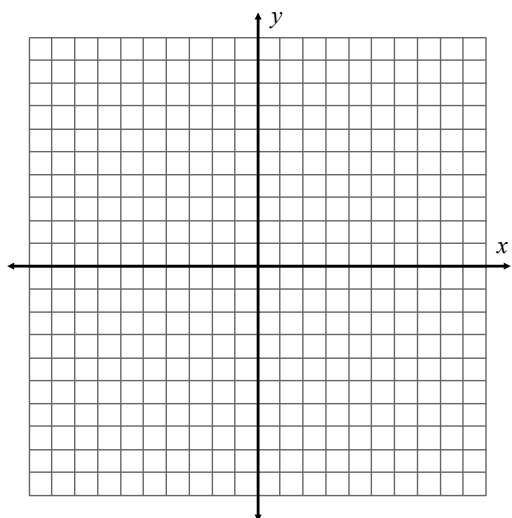
3) Triangle GHI has vertices $G(0,4)$, $H(6,0)$, and $I(2,-5)$. Find the coordinates of the vertices of $G'H'I'$ after the reflection $r_{y\text{-axis}}$.



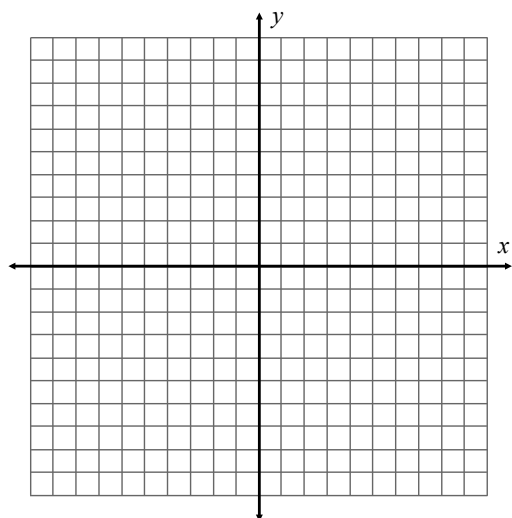
4) Triangle PQR has vertices $P(-6,9)$, $Q(2,5)$, and $R(-7,-1)$. Find the coordinates of the vertices of $P'Q'R'$ after the line $x = -3$.



5) Triangle ABC has vertices $A(5,-3)$, $B(9,0)$, and $C(7,5)$. Graph and label $\Delta A'B'C'$, the image of ΔABC under the transformation $r_{y=x}$.

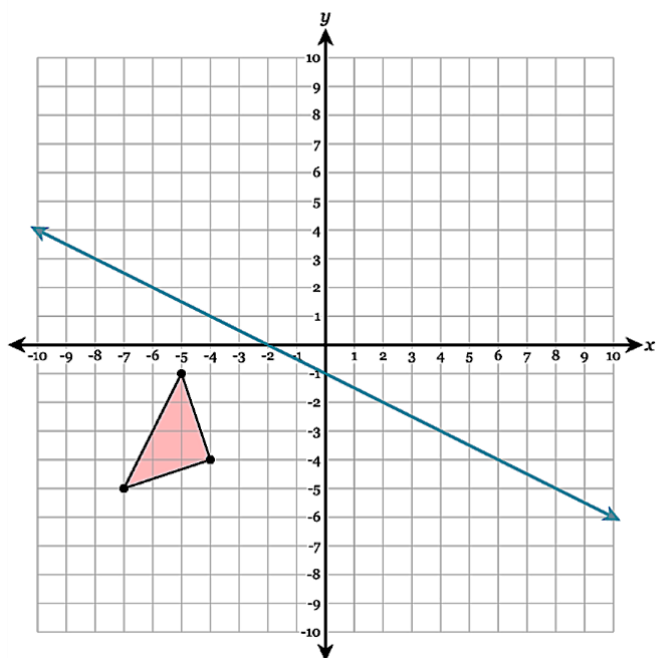


6) Triangle DEF has vertices $D(-3,1)$, $E(-5,5)$, and $F(0,0)$. Graph and label $\Delta D'E'F'$, the image of ΔDEF under the transformation $r_{y=-x}$.



Challenge:

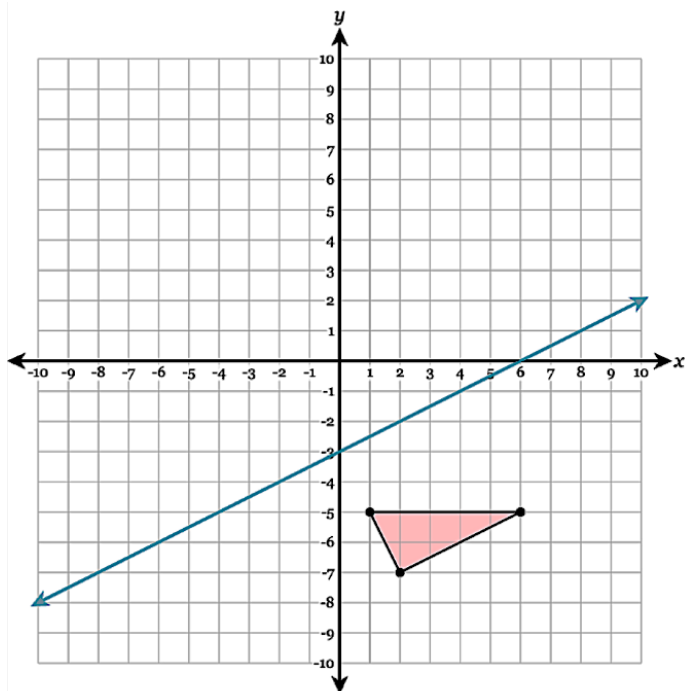
7) Reflect the image shown below over the line $y = -\frac{1}{2}x - 1$



$$m_1 =$$

$$m_{\perp} =$$

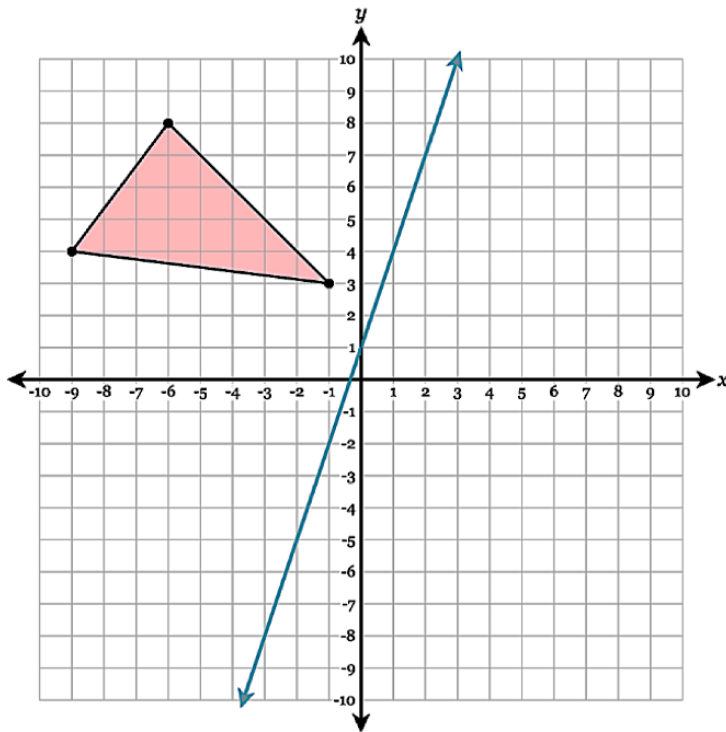
8) Reflect the image shown below over the line $y = \frac{1}{2}x - 3$



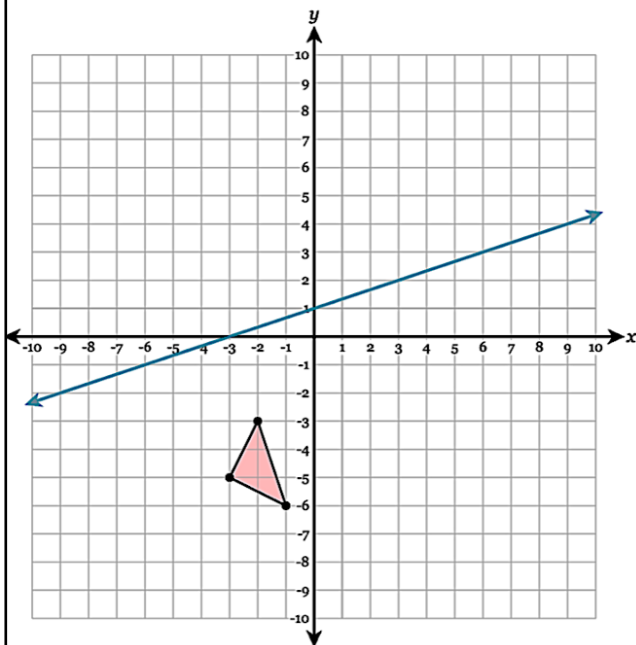
$$m_1 =$$

$$m_{\perp} =$$

9) Reflect the image shown below over the line $y = 3x + 1$

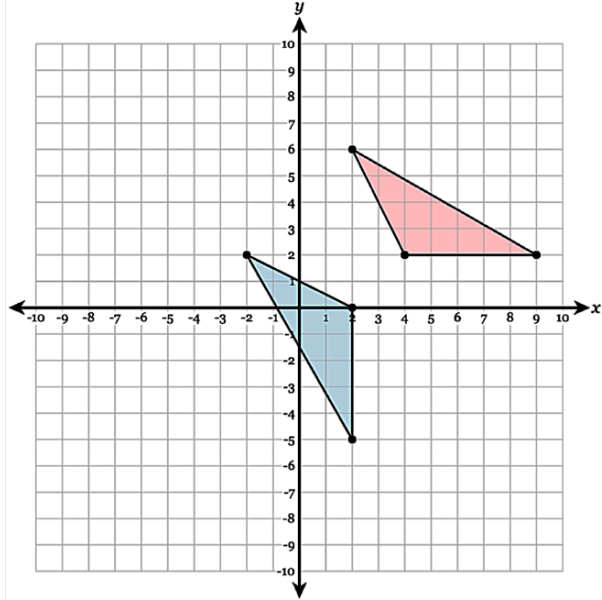


10) Reflect the image shown below over the given line.



Challenge

1) Determine the equation of the line of reflection.



Challenge

1) Determine the equation of the line of reflection.

